

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0248

Murray 138 kV
August 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0248. The Interconnection Customer proposes a 149.7 MW solar photovoltaic facility interconnecting at the Murray 138 kV substation in the New Jersey area. The request was received on June 28, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 184 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,430,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Heaton to Jarrett 138 KV L, the Crescentville to Burholme 138 KV Line loads to 141.3% of its 191 MVA rating. Project DFAX: 10.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crescentville to Burholme 138 KV Line	Rebuild line with higher-capacity conductor	\$1,430,000	100% (\$1,430,000)
	Total allocated to project		\$1,430,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.